Dark Energy as Inverse Holographic Information: A Parameter-Free $w(z)$ from an Informational Reading of the Cosmological Constant

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Abstract

The cosmological constant problem is usually stated as a mismatch of some 122 orders of magnitude between the Planck-scale vacuum energy density and the observed dark-energy density. Following a line of thought initiated by Banks and developed by Padmanabhan, we first exhibit the exact algebraic identity $\rho_\Lambda = \frac{3}{8}\,\rho_p/N_H$, where $N_H \equiv A_H/4\ell_p^2 \simeq 3.3\times10^{122}$ is the Bekenstein–Hawking information content of the de Sitter horizon associated with Λ. As it stands, this identity is a reformulation, not an explanation: it relocates the question "why is ρ_Λ small?" to "why does the accessible universe carry $\sim 10^{122}$ holographic degrees of freedom?". The contribution of this paper is to *promote* the identity from a kinematic statement about the asymptotic horizon to a dynamical law referred to the instantaneous future event horizon. The resulting model is shown to coincide exactly with Li's holographic dark energy, but with the usually free parameter fixed to $c = 1$ by the informational normalization (one bit per $4\ell_p^2$). The model therefore makes a genuinely parameter-free prediction for the dark-energy equation of state: $w_0 = -\frac{1}{3} - \frac{2}{3}\sqrt{\Omega_{\Lambda 0}} \simeq -0.885$, within 1σ of the DESI DR2+CMB+PantheonPlus measurement $w_0 = -0.838 \pm 0.055$, together with a definite thawing history $w(z) \to -\frac{1}{3}$ at high redshift and an asymptotic de Sitter future. The predicted sign of the evolution, however, is opposite to the central CPL fit of DESI ($w_a > 0$ here versus $w_a < 0$ there), which makes the model sharply falsifiable by forthcoming data. We close with a methodological analysis of what is, and what is not, explained by informational reformulations of Λ, and of the epistemic step that converts an accounting identity into empirical physics.

Keywords: cosmological constant problem; holographic principle; dark energy; equation of state; DESI; philosophy of cosmology.

1 Introduction

The cosmological constant problem is the observation that the vacuum energy density suggested by quantum field theory with a Planck-scale cutoff exceeds the observed dark-energy density by roughly 122 orders of magnitude (Weinberg, 1989; Martin, 2012). Within ΛCDM this mismatch is treated as an extreme cancellation to be explained by unknown ultraviolet physics.

Two developments motivate revisiting the problem from an informational standpoint. First, the holographic principle (Bekenstein, 1973; Hawking, 1975; 't Hooft, 1993; Susskind, 1995) teaches that the information content of a causal region is measured by its boundary area in Planck units, $S = A/4\ell_p^2$; applied to cosmological horizons this assigns the observable universe a finite information budget of order 10^{122}–10^{123} (Lloyd, 2002; Egan and Lineweaver, 2010). Banks

(Banks, 2000) proposed long ago that the logical order be inverted: Λ is not a small energy to be explained but an *input* that fixes the dimension of the Hilbert space of an asymptotically de Sitter universe; Padmanabhan and Padmanabhan (Padmanabhan and Padmanabhan, 2013; Padmanabhan, 2012) similarly argued that the numerical value of Λ is controlled by the cosmic information content ("CosMIn"). Second, and quite recently, the DESI collaboration has reported growing evidence—at the 2.8–4.2σ level depending on the supernova compilation—that dark energy is *not* a constant: the equation of state today satisfies $w_0 > -1$ and evolves with redshift (DESI Collaboration, 2024, 2025; Lodha et al., 2025). If dark energy is dynamical, frameworks in which ρ_Λ is tied to an evolving horizon quantity become natural candidates, whereas a strictly constant Λ cannot accommodate the data.

This paper does three things. Section 2 states the informational identity in its exact form and is deliberately candid about its epistemic status: it is an algebraic reformulation of known relations, not a derivation of Λ. Section 3 performs the step that gives the reformulation empirical content: the identity, referred at each cosmic time to the instantaneous future event horizon, becomes a dynamical law. We show that this law is precisely Li's holographic dark energy (Cohen et al., 1999; Li, 2004; Wang et al., 2017) with the normally free parameter fixed to $c = 1$, so that the entire $w(z)$ history follows with *no adjustable parameters* beyond $\Omega_{\Lambda 0}$. Section 4 derives the predictions and confronts them with DESI DR2. Section 5 collects the known difficulties of the framework, and Section 6 offers a methodological analysis of the move from interpretation to prediction. Throughout, $\ell_p^2 = \hbar G/c^3$, $\rho_p = c^5/\hbar G^2$, and $\rho_\Lambda = \Lambda c^2/8\pi G$ (mass densities).

2 The informational identity and its epistemic status

For a de Sitter horizon of radius $R_H = \sqrt{3/\Lambda}$, the horizon area is $A_H = 4\pi R_H^2 = 12\pi/\Lambda$, and the Bekenstein–Hawking relation assigns it the dimensionless information content

$$N_H \equiv \frac{A_H}{4\ell_p^2} = \frac{3\pi c^3}{\Lambda \hbar G} \approx 3.3 \times 10^{122}, \tag{1}$$

where the numerical value uses $\Lambda \approx 1.09 \times 10^{-52}\,\mathrm{m}^{-2}$ from current data (Planck Collaboration, 2020; DESI Collaboration, 2025). This is the familiar de Sitter entropy (Gibbons and Hawking, 1977); it agrees in order of magnitude with Lloyd's count of the computational capacity of the universe ($\sim 10^{120}$ elementary operations) (Lloyd, 2002) and with the entropy budget of Egan and Lineweaver (Egan and Lineweaver, 2010).

Combining Eq. (1) with the definitions of ρ_p and ρ_Λ gives the exact identity

$$\boxed{\rho_\Lambda = \frac{3}{8}\frac{\rho_p}{N_H}} \tag{2}$$

The famous factor 10^{-122} is thus, numerically, the inverse holographic information content of the Λ-horizon.

Three remarks fix the epistemic status of Eq. (2).

(i) It is not a derivation of Λ. Both sides are functions of the same measured Λ; the relation is an algebraic identity within de Sitter geometry and has, by itself, no predictive content. Any claim to have "derived" the dark-energy density from Eq. (2) would be circular.

(ii) It is nevertheless a nontrivial reframing. As Banks emphasized (Banks, 2000), one may read the identity right-to-left: the small number is not a fine-tuned cancellation but the reciprocal of a large integer—the number of holographic degrees of freedom of the accessible universe. The explanatory burden is relocated from ultraviolet cancellation mechanisms to the question of why the cosmic Hilbert space has the dimension it has (Padmanabhan and

Padmanabhan, 2013). Relocation is not resolution, but it can redirect research: questions about horizon information admit dynamical treatment in a way that questions about exact cancellation of 10^{122} contributions do not.

(iii) A caveat about which horizon. Today's universe is not exactly de Sitter ($\Omega_{\Lambda 0} \simeq 0.7$); the Hubble radius c/H_0 and the asymptotic radius $\sqrt{3/\Lambda}$ differ by a factor $\sqrt{\Omega_{\Lambda 0}}$. Equation (2) strictly refers to the *final* de Sitter horizon determined by Λ itself, which is the formal root of the circularity noted in (i). The next section removes this circularity by referring the information content to a horizon defined by the expansion history rather than by Λ.

3 From interpretation to dynamics: the promotion postulate

The step that converts Eq. (2) from an accounting identity into physics is the following postulate.

Postulate (informational dark energy). *At every cosmic time t, the dark-energy density equals $\frac{3}{8}\rho_p/N_H(t)$, where $N_H(t) = A_h(t)/4\ell_p^2$ is the Bekenstein–Hawking information content of the instantaneous future event horizon*

$$R_h(t) \;=\; a(t)\int_t^\infty \frac{c\,dt'}{a(t')}. \tag{3}$$

Substituting $N_H(t) = \pi R_h^2/\ell_p^2$ into the postulate gives

$$\rho_\Lambda(t) \;=\; \frac{3}{8}\,\frac{\rho_p\,\ell_p^2}{\pi R_h^2(t)} \;=\; \frac{3c^2}{8\pi G}\,\frac{1}{R_h^2(t)}, \tag{4}$$

using $\rho_p\ell_p^2 = c^2/G$. Equation (4) is recognizable: it is exactly the holographic dark energy (HDE) density of Cohen, Kaplan and Nelson (Cohen et al., 1999) with Li's choice of the future event horizon as the infrared cutoff (Li, 2004), $\rho_{\rm HDE} = 3c_{\rm H}^2 M_p^2/R_h^2$ (with $M_p^{-2} = 8\pi G$, $\hbar = c = 1$), *with the usually free dimensionless parameter fixed to* $c_{\rm H} = 1$.

This identification cuts both ways and both directions matter. In one direction, it is a candor requirement: the dynamics explored below is not new—HDE has a twenty-year literature, including extensive observational fits (Li, 2004; Huang and Li, 2004; Wang et al., 2017). In the other direction, the informational reading adds something HDE by itself lacks: a *reason* for the normalization. In standard HDE, $c_{\rm H}$ is a free parameter fitted to data (fits typically prefer $c_{\rm H} \lesssim 1$ (Huang and Li, 2004; Wang et al., 2017)). Here $c_{\rm H} = 1$ is forced by the Bekenstein–Hawking count of exactly one bit per $4\ell_p^2$ of horizon area—the same factor of 1/4 that black-hole thermodynamics fixes (Bekenstein, 1973; Hawking, 1975). The model therefore has *zero* adjustable parameters in the dark sector: given $\Omega_{\Lambda 0}$, the entire expansion history is determined. From a methodological point of view (Section 6) this maximal rigidity is the model's chief virtue, since it makes the framework sharply falsifiable rather than flexibly accommodating.

For a spatially flat universe containing matter and the component (4), defining $\Omega_\Lambda = \rho_\Lambda/(3M_p^2H^2)$ and $x = \ln a$, the standard HDE derivation (Li, 2004) yields the autonomous equation

$$\frac{d\Omega_\Lambda}{dx} = \Omega_\Lambda\,(1-\Omega_\Lambda)\left(1+2\sqrt{\Omega_\Lambda}\right), \tag{5}$$

and the equation of state

$$w(z) \;=\; -\frac{1}{3} \;-\; \frac{2}{3}\sqrt{\Omega_\Lambda(z)}. \tag{6}$$

A self-contained derivation of Eqs. (5)–(6) is given in Appendix A. Because Eq. (6) confines w to the open interval $(-1, -\frac{1}{3})$ at all finite times, the model structurally forbids both a constant $w = -1$ and any phantom regime.

Table 1: Predicted history of the informational dark-energy model ($\Omega_{\Lambda 0} = 0.685$; no adjustable parameters).

z	$\Omega_\Lambda(z)$	$w(z)$
0.00	0.685	-0.885
0.25	0.551	-0.828
0.50	0.441	-0.776
1.00	0.295	-0.696
2.00	0.160	-0.600
3.00	0.104	-0.549
10.0	0.026	-0.441
∞	0	$-1/3$

4 Predictions and confrontation with DESI DR2

4.1 Parameter-free numbers

Evaluating Eq. (6) today with $\Omega_{\Lambda 0} = 0.685$ (Planck Collaboration, 2020) gives the headline prediction

$$w_0 \;=\; -\frac{1}{3} - \frac{2}{3}\sqrt{0.685} \;=\; -0.885. \tag{7}$$

No parameter has been tuned to obtain this number. For comparison, the DESI DR2 combination BAO+CMB+PantheonPlus yields $w_0 = -0.838 \pm 0.055$ (DESI Collaboration, 2025); the prediction (7) sits within 0.9σ of the measured central value. The same combination with the DES-Y5 supernova sample gives $w_0 = -0.752 \pm 0.057$, about 2.3σ from (7).

Integrating Eq. (5) from $\Omega_{\Lambda 0} = 0.685$ gives the full history (Table 1, Fig. 1). The model is a *thawing-to-de-Sitter* scenario: $w \to -\frac{1}{3}$ at high redshift, $w_0 = -0.885$ today, and $w \to -1$, $\Omega_\Lambda \to 1$ in the far future. There is no phantom regime and no big rip; the future is asymptotically de Sitter, closing the loop with the static identity of Section 2, which becomes exact in that limit. The early dark-energy fraction is negligible where it must be: extrapolating the matter-plus-dark-energy system (radiation neglected, so the number is indicative only) gives $\Omega_\Lambda(z{=}1100) \sim 2 \times 10^{-4}$, far below current CMB bounds on early dark energy, which sit at the percent level.

4.2 Agreement, tension, and falsifiability

The confrontation with DESI must be reported symmetrically.

Points of agreement. The model predicts, unavoidably, that (a) dark energy is dynamical, so that ΛCDM is excluded within the framework; (b) $w_0 > -1$ today, i.e. quintessence-like present behavior; and (c) a present value $w_0 = -0.885$ within 1σ of the DESI+CMB+PantheonPlus measurement. Points (a) and (b) are precisely the two qualitative headlines of the DESI results (DESI Collaboration, 2024, 2025). It is worth emphasizing how nontrivial (c) is: a formula with no tunable constants, whose only inputs are the Bekenstein–Hawking factor 1/4 and $\Omega_{\Lambda 0}$, lands within one standard deviation of the measured w_0.

Point of tension. Projected onto the CPL parametrization $w(a) = w_0 + w_a(1 - a)$ over $0 < z < 1$, the model gives an effective $w_a \simeq +0.4$ (locally $+0.23$ at $z = 0$): w *increases* into the past toward $-\frac{1}{3}$. The DESI CPL fits prefer the opposite sign, $w_a \simeq -0.6$ to -1, i.e. a phantom regime $w < -1$ at $z \gtrsim 0.5$ (DESI Collaboration, 2025). Taken at face value this is a $\sim 4\sigma$ tension in w_a (a face-value statement only, since the DESI w_0–w_a posteriors are strongly correlated and a proper comparison must be made against the joint contour). Three qualifications are in order.

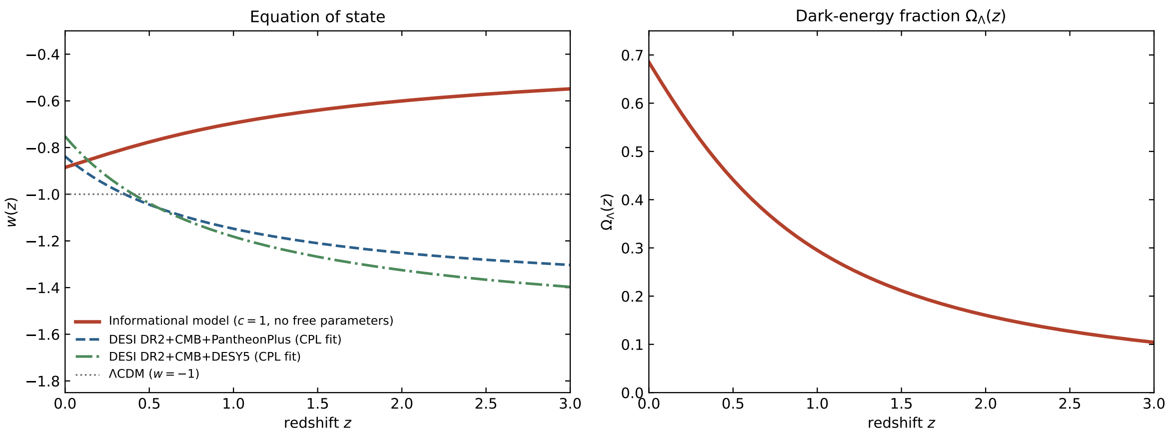

Figure 1: Left: equation of state $w(z)$ of the informational model (solid), compared with the CPL best fits of DESI DR2 combined with CMB and two supernova samples (DESI Collaboration, 2025), and with ΛCDM. Right: the predicted dark-energy fraction $\Omega_\Lambda(z)$. The model has no free parameters once $\Omega_{\Lambda 0}$ is fixed.

First, the CPL fit is a two-parameter extrapolation, and its high-redshift behavior is dominated by the parametrization rather than by data; model-independent reconstructions of $w(z)$ carry large uncertainties beyond $z \sim 0.7$, and the literature cautions explicitly against over-reading CPL extrapolations (Nesseris et al., 2025; Lodha et al., 2025). Second, a proper test of the present model is a direct fit of its distance–redshift relation to the BAO and supernova data, not a comparison of CPL coefficients; this fit (which involves no free dark-sector parameters, hence no ability to hide) is the natural next step and is left for dedicated work. Third, precisely because nothing can be adjusted, the model is *maximally exposed*: if forthcoming DESI and supernova data confirm a genuine phantom crossing at $z \approx 0.5$, the model is refuted outright. We regard this exposure as a feature. A framework born as an interpretation has become an empirical claim with a definite kill condition.

5 Known challenges

Honesty requires listing the difficulties this framework inherits and adds.

The future event horizon and causality. The cutoff (3) refers to the entire future expansion history, so the present dark-energy density "knows about" the future. This teleological flavor is a standing objection to event-horizon HDE (Cai, 2007; Wang et al., 2017). Two readings are available: as a genuine causality violation (fatal), or as a global self-consistency condition—the universe's history solves a boundary-value problem rather than an initial-value problem, a reading with precedents in action-at-a-distance electrodynamics and in global formulations of quantum cosmology. Within the informational framework the second reading is at least natural: horizon information is a global, not local, bookkeeping. But we flag it clearly as the framework's deepest conceptual liability.

Why one bit per $4\ell_p^2$ of the cosmological horizon? The Bekenstein–Hawking normalization is established for black-hole horizons and strongly motivated for de Sitter space (Gibbons and Hawking, 1977); its application to the time-dependent event horizon of an FRW universe is an extrapolation. The Cohen–Kaplan–Nelson bound (Cohen et al., 1999) motivates $\rho_\Lambda \lesssim M_p^2/L^2$ as an inequality; promoting it to an equality with coefficient fixed by horizon thermodynamics is the physical content of our postulate, and it could be wrong in either the coefficient or the choice of surface. (Choosing the Hubble radius instead is known to fail to produce acceleration

(Hsu, 2004), which is itself evidence that the choice of surface carries physical content.)

Relation to observationally preferred HDE. Fits of standard HDE with $c_{\rm H}$ free tend to prefer $c_{\rm H} < 1$ (Huang and Li, 2004; Wang et al., 2017), which entails phantom behavior in the future. Our normalization forbids this region. Should data ultimately demand $c_{\rm H} < 1$, the informational normalization—not merely the model—would be falsified.

What N_H counts. N_H is a capacity (a bound on accessible degrees of freedom), not a count of particles or bits "in use". The framework does not yet supply a microscopic account of what carries the horizon information, nor of how matter degrees of freedom are subsumed in it. Here the informational reading connects naturally to broader emergent-gravity programs (Padmanabhan, 2010; Verlinde, 2011), but the connection remains programmatic.

6 Methodological analysis: what has and has not been explained

The trajectory of this paper—identity, reinterpretation, promotion, prediction—is itself of philosophical interest, and we make the structure explicit.

Interpretations are cheap; rigid interpretations are not. The bare identity (2) has no empirical content: it cannot conflict with any observation, because both sides co-vary with the measured Λ. In Popperian terms it forbids nothing. What confers empirical content is the promotion postulate of Section 3, which extends the identity counterfactually—to all times, via a horizon defined independently of Λ. The promoted theory forbids a great deal: it forbids a constant $w = -1$ (the de Sitter value is attained only asymptotically), forbids phantom crossing, forbids any w_0 outside a narrow band around -0.89, and fixes the full function $w(z)$. The epistemic lesson generalizes: an interpretation earns scientific standing exactly when it is rigid enough to be killed. The absence of adjustable parameters, often presented apologetically, is on this view the model's principal epistemological asset.

Relocation of explanatory burden. Even if the model survives observation, one should be precise about what has been explained. The value of Λ (equivalently N_H, equivalently $\Omega_{\Lambda 0}$ at our epoch) remains an unexplained input, exactly as in Banks's proposal (Banks, 2000) and in CosMIn (Padmanabhan and Padmanabhan, 2013). What the framework explains, conditionally, is the *structure* of dark energy: why its density is tied to the horizon scale, why it evolves, and with what history. The fine-tuning question is transformed, not dissolved: "why is ρ_Λ small?" becomes "why is the cosmic information capacity large?"—a question that at least admits dynamical treatment, since horizon information grows with cosmic evolution and its magnitude is tied to the age and content of the universe (Lloyd, 2002; Egan and Lineweaver, 2010).

The Ptolemaic risk, faced honestly. A skeptic may object that identifying 10^{122} with a horizon-bit count is numerology—epicycles in informational dress. The reply is the one physics has always given: the difference between numerology and physics is a differential prediction. The static identity of Section 2 cannot be distinguished from coincidence; the dynamical model of Section 3 can, and DESI-class data are already adjudicating. Whichever way that adjudication goes, the methodological point stands: informational reformulations of Λ must be pushed to the point of rigidity at which they expose themselves to data, or they remain interpretive commentary.

7 Conclusion

We have separated, deliberately, three layers that are often run together in holographic discussions of the cosmological constant: an exact but non-predictive identity ($\rho_\Lambda = \frac{3}{8}\rho_p/N_H$); an interpretive inversion due to Banks and Padmanabhan (read Λ as fixing the cosmic information capacity, not as a fine-tuned energy); and a dynamical promotion (apply the identity at every epoch to the instantaneous event horizon). The third layer is equivalent to holographic dark energy with its free parameter frozen at $c_{\mathrm{H}} = 1$ by the Bekenstein–Hawking normalization, and therefore predicts, with no tunable constants, $w_0 = -0.885$—within 1σ of DESI DR2+CMB+PantheonPlus—together with a definite thawing history whose past behavior ($w_a > 0$) differs in sign from the current CPL central fit and will be cleanly tested by forthcoming data. A direct, parameter-free fit of the model's distance–redshift relation to DESI BAO and supernova compilations is the immediate next step.

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Data and code availability. No new observational data were produced. The short numerical script that integrates Eq. (5) and generates Table 1 and Fig. 1 is available from the author on request.

A Derivation of the autonomous equation and the equation of state

For completeness we derive Eqs. (5)–(6) from the postulate, following Li (2004). Units $\hbar = c = 1$, $M_p^{-2} = 8\pi G$; flat FRW with matter ($\rho_m \propto a^{-3}$) and the informational component $\rho_\Lambda = 3M_p^2/R_h^2$ (Eq. (4) with $c_{\mathrm{H}} = 1$).

From the definition (3) of the future event horizon,

$$\dot{R}_h \;=\; HR_h - 1. \tag{8}$$

The Friedmann equation $3M_p^2H^2 = \rho_m + \rho_\Lambda$ and the definition $\Omega_\Lambda \equiv \rho_\Lambda/3M_p^2H^2$ give

$$\Omega_\Lambda = \frac{1}{(HR_h)^2} \qquad \Longrightarrow \qquad HR_h = \Omega_\Lambda^{-1/2}. \tag{9}$$

With $x \equiv \ln a$ and $' \equiv d/dx = H^{-1}d/dt$, Eq. (8) gives

$$\frac{R_h'}{R_h} \;=\; 1 - \frac{1}{HR_h} \;=\; 1 - \sqrt{\Omega_\Lambda}. \tag{10}$$

Differentiating $\rho_\Lambda = 3M_p^2/R_h^2$ and comparing with the continuity equation $\rho_\Lambda' = -3(1+w)\rho_\Lambda$ yields the equation of state directly:

$$\rho_\Lambda' = -2\rho_\Lambda \frac{R_h'}{R_h} = -2\rho_\Lambda \left(1 - \sqrt{\Omega_\Lambda}\right) \;\;\Longrightarrow\;\; w = -\frac{1}{3} - \frac{2}{3}\sqrt{\Omega_\Lambda}, \tag{11}$$

which is Eq. (6). Since $0 < \Omega_\Lambda < 1$ at all finite times, w is confined to $(-1, -\frac{1}{3})$.

For the autonomous equation, differentiate the logarithm of Eq. (9),

$$\Omega_\Lambda'/\Omega_\Lambda = -2\left(H'/H + R_h'/R_h\right).$$

Differentiating the Friedmann equation and using $\rho_m' = -3\rho_m$ gives

$$\frac{H'}{H} = -\frac{3}{2}(1-\Omega_\Lambda) - \Omega_\Lambda\left(1 - \sqrt{\Omega_\Lambda}\right). \tag{12}$$

Combining the last three relations,

$$\frac{\Omega_\Lambda'}{\Omega_\Lambda} = 3(1-\Omega_\Lambda) + 2\Omega_\Lambda(1-\sqrt{\Omega_\Lambda}) - 2(1-\sqrt{\Omega_\Lambda}) = (1-\Omega_\Lambda)\left(1 + 2\sqrt{\Omega_\Lambda}\right), \tag{13}$$

which is Eq. (5). The fixed points are $\Omega_\Lambda = 0$ (unstable; matter domination, $w \to -\frac{1}{3}$) and $\Omega_\Lambda = 1$ (attractor; de Sitter, $w \to -1$), confirming the thawing-to-de-Sitter history quoted in the text.

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